

A Safety Score Is Not a Certificate: Calibration After Action Selection in Offline RL

Anonymous submission

Abstract

A learned safety score is not a deployment certificate. Offline safe reinforcement learning systems increasingly attach safety scores or residual-budget claims to actions selected from learned candidate banks. We study false safety claims among the actions actually issued after deployment selection, a target distinct from policy training, marginal calibration, or average constraint satisfaction. Candidate generation, reward ranking, support filtering, and fallback logic can concentrate rare calibration failures on the selected action. We formalize selective safety certification over fixed query blocks, give finite-family audit guarantees for issued-claim risk and episode-budgeted claims, and instantiate the protocol as support-aware ACCS. On trained CAPSIQL, CPQ, and COptiDICE proposers on official DSRL CarCircle, top-action false certification reaches 7.17% and 8.30% in the high-signal CPQ/COptiDICE rows, while ACCS-v0 reduces issued false certification to 2.25%/2.17% with 80.18%/78.33% yield. A pre-registered fixed-cap closed-loop frontier separates proposer-environment pairs: CPQ certifies full CarCircle and BallCircle episodes with exact 95% episode-risk upper bounds of 3.22% and 0.50%, whereas COptiDICE reaches 64.15% and 90.89%. These results make post-selection auditing a certificate evidence layer: it can certify particular issued-claim contracts and reject unsafe or unsupported contracts, but it is not a replacement for safe policy learning.

Introduction

Offline safe reinforcement learning is increasingly used as a decision layer for systems that must trade reward against state-dependent safety budgets. Modern methods learn cost-aware critics, budget-conditioned policies, and constraint-adaptive action filters for offline safe RL [Li et al.(2025), Wang et al.(2026)]. Other recent directions build almost-sure safety or flow-policy formulations [Lee, Kwon, and Oh(2026), Tayal, Tayal, and Prakash(2026)]. These components are useful for choosing actions, but they do not by themselves answer a different statistical question: when the deployed system issues a safety claim for its selected action, how often is that claim false?

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The distinction matters because deployment systems do not certify random candidate actions. They generate a bank of candidates, filter by a requested residual budget, rank by reward, apply fallback rules, and often repeat this process under changing constraints. Marginal calibration over candidate proposals can therefore leave the selected action miscalibrated. In a simple construction, each proposed action has only 4% false-certification probability, but selecting the best of 16 reward-ranked candidates raises the issued-claim false-certification rate to about 48%. Calibration before action selection is not calibration after action selection.

This paper studies the missing evidence object: false certification among issued action claims. We define a deployment query block containing the state, residual budget, candidate action bank, learned reward and safety scores, support evidence, and residual-cost labels under a declared continuation policy. This target is related to selective conformal and conformal risk-control ideas [Bao et al.(2025), Angelopoulos et al.(2025)], but the RL setting adds candidate selection, support, and continuation-policy dependence. A certification rule either issues one action certificate or abstains. The target risk is the ratio of expected false issued claims to expected issued claims, and it must be reported together with yield to avoid vacuous safety by abstention.

This formulation changes what a valid evaluation must hold fixed. If an auditor is tested on easier candidates than the deployed proposer emits, low risk says little about deployment. If residual costs are labeled under one continuation policy but deployed under another, the certificate target has changed. If a rule issues on actions outside logged support, no distribution-free offline calibration can identify the missing residual outcomes. We therefore treat the query law, residual-label protocol, and support evidence as part of the certification problem rather than implementation details.

Our method is a support-aware selective auditor over fixed query blocks. The auditor compares a prespecified family of certification rules using independent audit blocks, selects rules whose upper confidence bound on issued-claim risk is below the target, and breaks ties by utility or yield. This design treats support as evidence rather than decoration: unsupported actions, policy-mismatched residual labels, and low-yield global thresholds are all diagnosed rather than silently certified.

The main line is that a certificate must be audited after

action selection under a frozen deployment contract. Around that line, we make three contributions. First, we isolate selected safety certification as an estimand for offline safe RL and prove that marginal candidate calibration can fail after reward-based action selection. Second, we give a finite-family audit protocol that controls issued-claim risk with explicit yield, support, and residual-policy accounting. Third, we build an empirical evidence stack from action-level query blocks to logged episode proxies and true closed-loop fixed-cap audits, making clear where action certificates do and do not imply trajectory-level claims.

The empirical result is already visible on trained safe-RL proposers, not only on exact toys. On official DSRL CarCircle, we audit CAPSIQL, CPQ, and COptiDICE checkpoint proposers under the same query-bank and residual-label contract [Liu et al.(2023), Li et al.(2025)]. Global candidate conformal prediction reaches low risk mostly by issuing very few certificates. Reward-bin and CRC-style auditors are strong baselines. ACCS-v0 keeps selected-claim risk near 2.2–2.3% while preserving about 78–81% yield in the high-signal CPQ and COptiDICE rows. The closed-loop fixed-cap frontier then tests a stricter episode-level question: CPQ stays below the 5% episode-risk target when the fixed cap saturates full CarCircle and BallCircle episodes, whereas COptiDICE fails sharply and DroneCircle exposes a harder residual-risk boundary. This layered result is the paper’s scope: post-selection auditing is a certificate evidence layer, not a claim that every safe-RL policy is safe online.

Related Work

Offline safe reinforcement learning provides the action-generation substrate for our audit problem. DSRL and Safety-Gymnasium define benchmark environments and logged datasets for reward-cost trade-offs [Liu et al.(2023), Ji et al.(2023)]. CAPS, budget-conditioned reachability, almost-sure safety, robust shielding, and SafeFQL-style flow policies train or select actions under safety constraints [Li et al.(2025), Wang et al.(2026), Lee, Kwon, and Oh(2026), Anonymous(2026), Tayal, Tayal, and Prakash(2026)]. These methods address how to build a safer proposer. We instead ask whether the certificate attached to the proposer-selected action remains valid after within-state candidate generation, filtering, ranking, and fallback.

Conformal safety methods calibrate prediction sets, safety filters, or off-policy value estimates for control systems [Selig et al.(2023), Anonymous(2025), Foffano, Russo, and Proutiere(2023)]. Their guarantees are valuable but depend on the declared coverage target and data-generating law. In offline RL, the residual cost attached to an action is continuation-policy dependent, and unsupported actions are not identifiable from logs alone. ACCS therefore treats the residual-label protocol and support evidence as part of the query block rather than as implementation details.

Selective conformal prediction and conformal risk control are the closest statistical relatives [Bao et al.(2025), Angelopoulos et al.(2025)]. We do not claim that post-selection risk control is a new statistical phenomenon. The

contribution is the RL-specific instantiation: issued action claims are produced from a candidate bank, safety is measured by residual budgets, support/no-overlap changes what can be certified, and utility is meaningful only through the risk-yield tradeoff after action selection. This is why our experiments compare against global CP, reward-bin conditioning, CRC-style thresholding, and support-aware ACCS on the same frozen query banks.

Problem Setup

We model deployment as a stream of query blocks rather than isolated action labels. A block is

$$W = (S, B, \{A_j, R_j, C_j, O_j, Z_j\}_{j=1}^K),$$

where S is the state, B is the requested residual budget, A_j is candidate action j , R_j is a learned reward score, C_j is a learned safety or cost score, O_j is support evidence, and Z_j is the residual cost after taking A_j and then following a declared continuation policy. The candidate bank may come from a trained offline safe-RL proposer, a budget-conditioned actor, or a sampled action generator.

A certification rule maps a query block to either one issued action or abstention:

$$h(W) \in \{1, \dots, K, \text{abstain}\}.$$

Let

$$I_h(W) = \mathbb{I}\{h(W) \neq \text{abstain}\}$$

indicate that the rule issues a certificate, and let

$$F_h(W) = \mathbb{I}\{h(W) = j \text{ and } Z_j > B\}$$

indicate that the issued certificate is false. The main evaluation target is

$$\rho(h) = \frac{\mathbb{E}[F_h(W)]}{\mathbb{E}[I_h(W)]},$$

paired with yield(h) = $\mathbb{E}[I_h(W)]$. A method is useful only if it controls issued-claim risk while maintaining nontrivial yield and utility.

This target is different from ordinary marginal conformal coverage. A bound U satisfying $\Pr[Z \leq U] \geq 1 - \alpha$ implies $\Pr[U \leq B, Z > B] \leq \alpha$, but it does not imply $\Pr[Z > B \mid U \leq B] \leq \alpha$. The conditional denominator is exactly the issued-claim yield. If certification is rare, or if selection concentrates the errors, selected-claim risk can be much larger than the marginal error rate. This also highlights why policy-specific residual labels and off-policy query laws must be stated explicitly [Barber et al.(2020), Foffano, Russo, and Proutiere(2023)].

Theory

The key theoretical point is that selected-action certification is a distinct statistical estimand. Marginal candidate calibration can be correct and still leave the issued action claim unreliable.

Theorem 1 (Selection amplification). *Fix $p \in (0, 1)$ and a candidate-bank size K . There is a one-state query law under which every candidate action has false-certification probability p , but the reward-selected action has issued-claim risk*

$$\rho(h_{\max}) = 1 - (1 - p)^K.$$

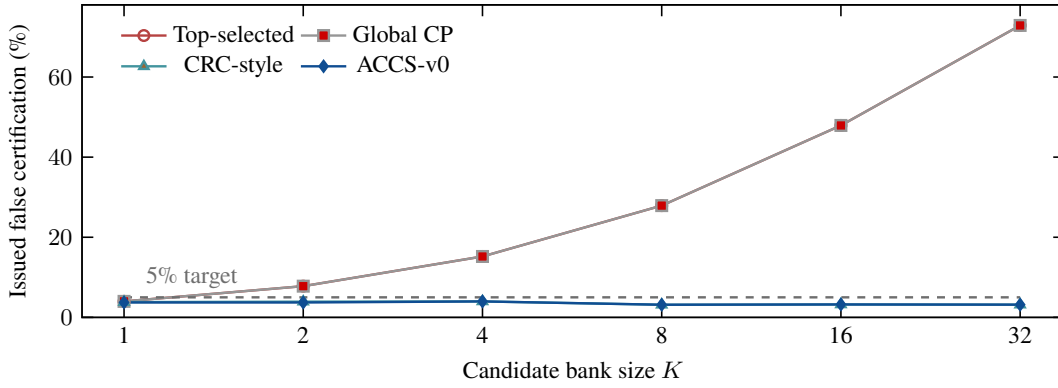


Figure 1: Selection can amplify false certification after marginal proposal-level calibration. In the exact toy, each candidate action has 4% false-certification probability, but selecting the highest-reward action from a candidate bank sharply increases issued-claim risk. Stronger post-selection baselines repair the toy, which motivates our focus on RL-specific query blocks, support, and trained proposer audits.

Proof sketch. Sample $X_1, \dots, X_K$ independently from $\text{Unif}[0, 1]$, set reward $R_j = X_j$, certify every candidate, and let a candidate be false when $X_j > 1 - p$. The deployed rule chooses $h_{\max}(W) = \arg \max_j R_j$. The issued claim is false exactly when at least one candidate lies in the failing tail, so the risk is $1 - \Pr[\max_j X_j \leq 1 - p] = 1 - (1 - p)^K$. For $p = 0.04$ and $K = 16$, this is approximately 0.48, matching Figure 1.

Theorem 2 (Finite-family selective-risk control). *Let $\mathcal{H} = \{h_1, \dots, h_M\}$ be fixed before observing audit labels, and let $W_1, \dots, W_n$ be independent audit query blocks from the target query law. For each rule define*

$$\hat{f}_h = \frac{1}{n} \sum_{i=1}^n F_h(W_i), \quad \hat{i}_h = \frac{1}{n} \sum_{i=1}^n I_h(W_i),$$

and $\epsilon_n = \sqrt{\log(4M/\delta)/(2n)}$. With probability at least $1 - \delta$, every rule satisfying

$$\hat{i}_h > \epsilon_n, \quad \frac{\hat{f}_h + \epsilon_n}{\hat{i}_h - \epsilon_n} \leq \alpha, \quad \hat{i}_h - \epsilon_n \geq \gamma$$

also satisfies $\rho(h) \leq \alpha$ and $\text{yield}(h) \geq \gamma$ under the same query law.

Proof sketch. Apply Hoeffding’s inequality to $F_h(W)$ and $I_h(W)$ for each fixed rule, then union bound over $2M$ empirical means. On the simultaneous good event, $\mathbb{E}[F_h] \leq \hat{f}_h + \epsilon_n$ and $\mathbb{E}[I_h] \geq \hat{i}_h - \epsilon_n$, so dividing the numerator upper bound by the positive denominator lower bound gives the result. This theorem justifies selecting over a prespecified support-, score-, and rank-threshold family using independent audit blocks.

Theorem 3 (Episode-budgeted selective-risk control). *Let $\mathcal{A} = \mathcal{H} \times \mathcal{M}$ be a finite family of base rules and per-episode emission budgets fixed before audit outcomes are observed. For an episode E , let $J_a(E)$ indicate that budgeted rule $a = (h, m)$ issues at least one certificate and let $G_a(E)$ indicate that at least one issued certificate is false. Given independent*

audit episodes $E_1, \dots, E_N$ and $\epsilon_N = \sqrt{\log(4|\mathcal{A}|/\delta)/(2N)}$, any pair satisfying

$$\hat{j}_a > \epsilon_N, \quad \frac{\hat{g}_a + \epsilon_N}{\hat{j}_a - \epsilon_N} \leq \alpha_{\text{ep}}, \quad \hat{j}_a - \epsilon_N \geq \gamma_{\text{ep}}$$

has episode selective risk $\mathbb{E}[G_a(E)]/\mathbb{E}[J_a(E)] \leq \alpha_{\text{ep}}$ and episode yield at least γ_{ep} with probability at least $1 - \delta$.

Proof sketch. Apply the previous finite-family argument to the episode-level indicators $G_a(E)$ and $J_a(E)$. The exchangeable unit is now the episode rather than the query block, and the emission budget is part of the prespecified rule family.

Corollary 1 (Exact audit for a fixed emission rule). *Fix an episode-emission rule a before observing audit labels. Conditional on the $N > 0$ audit episodes on which a issues at least one certificate, let K contain a false issued certificate. Then the one-sided Clopper–Pearson upper bound $U_{\text{CP}}(K, N, \delta)$ satisfies $\rho_{\text{ep}}(a) \leq U_{\text{CP}}(K, N, \delta)$ with probability at least $1 - \delta$.*

Proof sketch. For a fixed rule, issued audit episodes are iid from the conditional issued episode law, and their false-issued indicators are Bernoulli with mean $\rho_{\text{ep}}(a)$. The exact binomial confidence bound applies without a finite-family penalty. This is the route used by the fixed-cap closed-loop audit.

Proposition 1 (No-overlap impossibility). *Suppose the offline data distribution assigns zero probability to a deployment query region E , but the certifier may issue safe certificates on E with positive probability. Then there exist two CMDPs that induce identical offline data distributions and identical residual costs on supported actions, but for which the residual safety outcome on E is safe in one CMDP and unsafe in the other. Any data-dependent certifier that issues on E must falsely certify in one of these two worlds, conditional on those unsupported issued claims.*

The implication is methodological rather than merely conservative. Offline safe-RL certificates must state the continuation policy defining Z_j and must carry explicit support,

simulator, model, or weighting assumptions for the deployment query law. Unsupported high-reward actions are not a calibration problem; they are an identification problem.

Method

ACCS is a post-selection auditing protocol for a fixed proposer and query law. It does not retrain the policy and does not modify the candidate generator. Instead, it asks which certificates are statistically defensible for the actions the deployment rule is already prepared to issue. The protocol has three modules: freeze the query law, audit a finite family of certification rules on independent query blocks, and optionally add an emission budget when the claim is lifted from action blocks to episodes.

Freezing the Deployment Query Law

The first module freezes the evidence object before any auditor comparison. We fix the proposer checkpoint, candidate-bank size, residual budget distribution, residual-label horizon or simulator rollout, support score, and train/audit/test split. This prevents a subtle but serious evaluation error: an auditor can look safe simply because it is evaluated on easier candidate actions than the deployed rule would actually rank. Once the query blocks are frozen, every baseline receives the same candidate actions, reward scores, safety scores, support values, and residual-cost labels.

Residual labels are always tied to a declared continuation policy. In logged DSRL evaluations, Z_j is a suffix cost computed after replacing the logged action by the candidate-matched action and then following the logged suffix for a fixed horizon. In closed-loop evaluations, Z_j is the realized online episode cost after deploying the trained proposer. These two labels answer different questions, so the paper reports them as different evidence layers rather than merging them into one safety number.

Auditing Fixed Certification Rules

The second module estimates issued-claim risk for a finite rule family. For each $h \in \mathcal{H}$ and audit blocks $W_1, \dots, W_n$, we compute

$$\hat{f}_h = \frac{1}{n} \sum_{i=1}^n F_h(W_i), \quad \hat{i}_h = \frac{1}{n} \sum_{i=1}^n I_h(W_i).$$

The empirical ratio $\hat{f}_h/\hat{i}_h$ is not enough, because low-yield rules have unstable denominators. ACCS therefore uses confidence bounds on both false-issued mass and issued mass, producing a conservative upper bound

$$\rho_{\text{UCB}}(h) = \frac{\hat{f}_h + \epsilon_h}{\max(\hat{i}_h - \epsilon_h, \epsilon_0)}.$$

ACCS selects a useful certificate rule rather than the safest vacuous rule. It keeps only rules satisfying $\rho_{\text{UCB}}(h) \leq \alpha$ and $\hat{i}_h - \epsilon_h \geq \gamma$, then chooses the remaining rule with the highest audit utility. In our current implementation, the rule family includes thresholds over support, learned safety score, reward rank, and fallback logic. This makes the method comparable to reward-bin and CRC-style baselines while preserving the support-aware abstention behavior needed for offline RL.

Algorithm 1 ACCS audit for fixed query blocks

Require: proposer π , query law $\mathcal{Q}$, rule family $\mathcal{H}$, audit blocks $\mathcal{D}_{\text{aud}}$, risk target α , yield floor γ

- 1: Generate and freeze query blocks $W = (S, B, \{A_j, R_j, C_j, O_j, Z_j\}_{j=1}^K)$ from π and $\mathcal{Q}$.
- 2: **for** each rule $h \in \mathcal{H}$ **do**
- 3: Compute issued indicators $I_h(W_i)$ and false-issued indicators $F_h(W_i)$ on $\mathcal{D}_{\text{aud}}$.
- 4: Compute $\rho_{\text{UCB}}(h)$ and a lower confidence bound on yield.
- 5: **end for**
- 6: Keep rules with $\rho_{\text{UCB}}(h) \leq \alpha$ and yield lower bound at least γ .
- 7: Return the feasible rule with highest audit utility, or abstain from certifying if no feasible rule exists.

Rule Family and Baselines

The rule family is deliberately simple and fixed before test labels are used. Rules may threshold the learned safety score, require a minimum support value, limit reward rank, or combine these conditions with a deterministic fallback. This design makes ACCS comparable to strong post-selection baselines. Global candidate conformal prediction learns one safety threshold before deployment. Reward-bin UCB conditions the audit on reward-score bins, directly addressing the fact that reward ranking can concentrate failures. CRC-style thresholding chooses a safety threshold by expected-risk control. ACCS-v0 differs by jointly using support and safety evidence under the finite-family risk-yield selection rule above.

The same audit spine supports a stricter episode-budgeted extension. For an episode, a budgeted rule applies the selected action-level auditor but issues at most m certificates in episode order. The audit target then becomes whether an issued episode contains any false certificate, not the fraction of false issued blocks. In logged data we can evaluate this as an episode-block proxy by grouping fixed query blocks by anchor episode; in simulator rollouts the same rule can be evaluated online. The distinction matters: the proxy is a risk-allocation diagnostic, while a closed-loop result additionally depends on the online state distribution induced by the deployed policy.

For the online fixed-cap audit, we use an even cleaner fixed-rule specialization. The rule is chosen before seeing audit labels: issue certificates for the first $m \in \{64, 128, 256, 512\}$ policy steps of each episode. Because no score threshold or rule choice is fit on audit outcomes, exact one-sided binomial confidence intervals over issued episodes are sufficient. This is why the closed-loop table reports Clopper–Pearson upper bounds rather than the larger finite-family UCB used for adaptive rule selection.

Experiments

The experiments are organized around three increasingly strict questions. The main benchmark asks whether an auditor controls selected false certification on fixed action query blocks. The episode-proxy diagnostic asks how those block

claims accumulate when many certificates are issued in one logged episode. The closed-loop audit asks whether a pre-registered sparse emission rule remains safe under online simulator rollouts. This separation is intentional: each layer has a different exchangeability unit and a different claim.

We evaluate action-level certification on fixed deployment query blocks rather than on individual logged transitions. For each test state and residual budget, a trained proposer generates a candidate action bank with reward, safety, and support scores. The deployment rule then selects or abstains on one issued action claim. All auditors are evaluated on the same frozen query banks, so differences come from certification rules rather than from changing the proposer distribution. We report false issued certification and yield together; an auditor that controls risk only by issuing almost no certificates is not a practical deployment rule.

The auditor suite separates proposal-level calibration from post-selection certification. The unaudited row selects the top-reward action from the candidate bank. Global candidate conformal prediction calibrates one threshold before selection. Reward-bin UCB conditions on the deployment reward score, and CRC-style thresholding selects a safety threshold by expected-risk control. ACCS-v0 adds support-aware selection over a prespecified finite rule family. We keep reward-bin and CRC-style auditors as strong baselines because local stress tests show that simple post-selection conditioning can repair some selection-amplification failures [Angelopoulos et al.(2025)].

All main action-level rows use the same official DSRL CarCircle dataset and the same audit/test split size. CAPSIQL, CPQ, and COptiDICE are trained or loaded as proposer checkpoints, then converted into query blocks by sampling candidate actions and matching them to official DSRL residual labels. We use three seeds for every promoted table row. The reported standard deviations are across checkpoint/query seeds, not across bootstrapped query blocks.

Table 1 evaluates three trained offline safe-RL proposers on official DSRL `OfflineCarCircle-v0`: CAPSIQL at 50k update steps, CPQ at 10k update steps, and COptiDICE at 10k update steps. CAPSIQL q0.93 is the current trained CAPS-style mainline: candidate false certification is 4.42%, but top-reward selection raises the issued-claim false rate to 5.68%. CPQ q0.92 and COptiDICE q0.92 are the high-signal direct OSRL rows, with top-selected false-certification rates of 7.17% and 8.30%.

Post-selection auditing reduces issued-claim risk while preserving substantially more yield than global calibration. In the CPQ high-signal row, ACCS-v0 reduces selected false certification from 7.17% to 2.25% while keeping 80.18% yield. In the COptiDICE stress row, ACCS-v0 reduces selected false certification from 8.30% to 2.17% with 78.33% yield. Global candidate conformal prediction has lower risk, but it issues certificates on only 7.33% and 8.08% of these two query blocks, respectively. The relevant comparison is therefore a risk-yield tradeoff, not risk alone.

We also stress the residual-label horizon on CPQ and COptiDICE at $q = 0.92$. Changing the logged suffix label from $H = 40$ to $H = 20$ keeps the selected-risk gap visible and gives high-yield audited rules: ACCS-v0 reaches 2.26% risk

at 94.82% yield on CPQ and 2.13% risk at 93.00% yield on COptiDICE. The harder $H = 80$ setting is more revealing: ACCS-v0 remains below the 5% target with 3.61%/80.13% on CPQ and 3.27%/84.93% on COptiDICE, while global or reward-bin rules become much closer competitors and can exceed the target in the COptiDICE row. This supports the moving-residual-budget motivation without turning action certificates into closed-loop episode-safety claims.

To make that boundary concrete, we also group the same logged test query blocks by their anchor episodes. This episode-block proxy is not a simulator closed-loop guarantee, but it shows what is lost when action claims are aggregated over many opportunities. On CPQ q0.92, ACCS-v0 has 2.25% block-level selected false certification at 80.18% yield, yet 49.08% of issued logged episodes contain at least one false issued claim. On COptiDICE q0.92, the corresponding numbers are 2.17% block risk at 78.33% yield and 44.37% episode-proxy false certification. This diagnostic supports our scope statement: ACCS certifies selected action claims, while episode-level safety requires a separate risk allocation or closed-loop audit.

The emission-budget proxy shows that this next layer is not vacuous. Reusing the same ACCS rule but capping certificate emission to one logged query block per episode reduces episode-proxy false certification to 0.46% on CPQ and 0.69% on COptiDICE, with 100% episode yield but only 2.42% block yield. Increasing the cap to four raises block yield to 9.67% while keeping episode-proxy false certification at 2.53% on CPQ and 2.64% on COptiDICE. Cap eight begins to exceed the 5% target, identifying a concrete risk-allocation curve for trajectory-level certificates.

Finally, we ran true closed-loop simulator diagnostics with a fully pre-registered fixed-cap rule: issue certificates for the first m policy steps in each online episode, with $m \in \{64, 128, 256, 512\}$ and no score threshold fit on audit labels. The audit and test splits each contain 200 online episodes per seed, and we aggregate three seeds using exact one-sided binomial bounds over issued episodes. Table 2 shows the cap512 test split across CarCircle, BallCircle, and DroneCircle. CPQ meets the 5% episode-risk target on CarCircle and BallCircle while issuing certificates for the full 300-step and 200-step episodes, respectively. COptiDICE fails sharply on both positive environments, and DroneCircle exposes a hard environment where neither proposer supports a positive 5% closed-loop claim. A separate tune/audit/test rule search improves CPQ DroneCircle from 90/600 false issued episodes (U95 17.61%) to 32/395 (U95 10.73%), but it still misses the target; we keep it as appendix evidence that simple threshold/cap tuning is not enough.

The main direct-OSRL table certifies action claims under the constructed offline query law, while the closed-loop result certifies a fixed sparse emission rule under the induced online rollout distribution. Neither proves robustness under arbitrary online adaptation or safety for unsupported actions. The separate policy-mismatch and no-overlap experiments motivate these boundaries, and the next benchmark extension should test whether the same risk-yield pattern survives stronger public comparators such as runnable SafeFQL checkpoints.

Role	Proposer	q	Cand. F	Top F	Glob. R/Y	R-bin R/Y	CRC R/Y	ACCS R/Y
mainline	CAPSIQL 50k	0.93	4.42±0.43	5.68±0.25	0.09 / 3.81	3.59 / 71.71	2.67 / 83.52	2.28 / 80.60
high-signal	CPQ 10k	0.92	6.40±0.68	7.17±0.67	0.08 / 7.33	1.99 / 60.96	2.78 / 83.44	2.25 / 80.18
stress	COptiDICE 10k	0.92	6.36±0.57	8.30±0.76	0.13 / 8.08	3.12 / 57.94	2.28 / 79.15	2.17 / 78.33
stability	CPQ 10k	0.93	4.13±0.49	4.60±0.51	0.05 / 7.33	1.88 / 73.48	2.82 / 91.97	2.42 / 89.49
stability	COptiDICE 10k	0.93	4.16±0.45	5.86±0.29	0.03 / 8.08	3.10 / 70.88	2.69 / 86.07	2.20 / 83.12

Table 1: Direct OSRL trained-proposer audit on official DSRL `OfflineCarCircle-v0`. Values are mean percentages over three seeds. Cand. F is candidate false rate; Top F is unaudited top-selected false certification; R/Y columns report selected false certification / claim yield.

Env.	Proposer	Cap	Test K/N	U95	Issued
Car	CPQ	512	12/600	3.22	300.0
Car	COpt.	512	365/600	64.15	300.0
Ball	CPQ	512	0/600	0.50	200.0
Ball	COpt.	512	533/600	90.89	200.0
Drone	CPQ	512	90/600	17.61	49.5
Drone	COpt.	512	576/600	97.23	271.9

Table 2: Pre-registered fixed-cap closed-loop test audit. K/N counts false issued episodes over three 200-episode test seeds at cap512; U95 is the exact one-sided binomial 95% upper bound, in percent. Issued is mean issued certificates per online episode; cap512 saturates CarCircle and BallCircle.

Diagnostic Checks

The exact toy in Figure 1 is a phenomenon check, not the full empirical argument. It proves that action selection can turn a small marginal false-certification rate into a large issued-claim false rate, but it also reveals a reviewer risk: simple post-selection baselines can repair this one-state construction. Reward-bin, rank-bin, and CRC-style variants all perform well on the hardened toy. The benchmark question is therefore whether a frozen RL query law with support, residual budgets, and trained proposers still needs a support-aware selective audit.

The policy-mismatch diagnostic checks a different failure mode. If residual labels are generated under a behavior-policy continuation but deployment uses a different continuation policy, the certificate target changes even when the initial state and action look identical. In the local construction, behavior labels can make accepted actions appear safe while deployment-continuation labels fail. This motivates the explicit Z_j definition and the separation between logged suffix labels and closed-loop simulator labels.

The no-overlap diagnostic checks identifiability rather than calibration. When the offline data distribution has no support on a high-reward deployment region, no amount of marginal conformal calibration can infer whether that region is safe. A method can abstain, require external support evidence, or make a model-based assumption, but it cannot honestly issue a distribution-free offline certificate there. This is the practical reason ACCS treats support as a first-class audit variable rather than a secondary feature.

The off-policy query-shift diagnostic tests an audit-law P versus deployment-law Q mismatch. At strong tilt $a = 8$, un-

weighted P -audit has 33.90% false certification at 41.19% yield under Q ; known-ratio weighting matches the Q oracle at 4.45%/28.49%, and a support cap gives 0%/27.23%. The point is not benchmark dominance, but that query-law changes need weighting or support assumptions.

These diagnostics are falsification tests for the claim form: toy-only success is not enough; continuation mismatch certifies the wrong label; outside-support extrapolation is not distribution-free; shifted query laws need weighting or support assumptions; and DroneCircle acceptance would overstate closed-loop generality. The intended pattern is positive gains plus correct failures under stress.

Audit Accounting and Scope

The audit target is intentionally narrower than a universal safe-RL guarantee. An issued action certificate is a statistical claim about the residual-cost label attached to a particular query block. A closed-loop fixed-cap certificate is a claim about whether the first few issued certificates in an online episode include any false claim. These claims match what a deployment system can log, audit, and report; they do not assert that every future trajectory is safe.

This accounting also explains why global conformal calibration is not the paper’s endpoint. Global CP is a valid conservative baseline for candidate claims, and in several rows it reaches very low false-certification rates. However, it often does so by issuing on only about 4–8% of query blocks in the trained-proposer table. A safety evidence layer for deployment must expose this denominator. ACCS is useful when it moves the selected-claim risk below the target while keeping enough yield for the certificate to matter.

The strongest limitation is the fixed query law. The direct OSRL experiments freeze candidate generation, support scoring, residual labels, and the audit split before comparing auditors. If deployment changes its proposer, candidate-bank size, budget distribution, or continuation policy, the audit must be rerun or justified by a new weighting/support argument. The policy-mismatch and no-overlap diagnostics show why: residual certificates can fail for reasons ordinary calibration cannot repair.

SafeFQL and related conformal safety methods are important comparators but answer a different question unless evaluated under the same issued-claim protocol. SafeFQL calibrates a learned safety boundary for a flow-policy formulation; our protocol audits certificates after a deployment rule has generated and selected among candidate actions. A di-

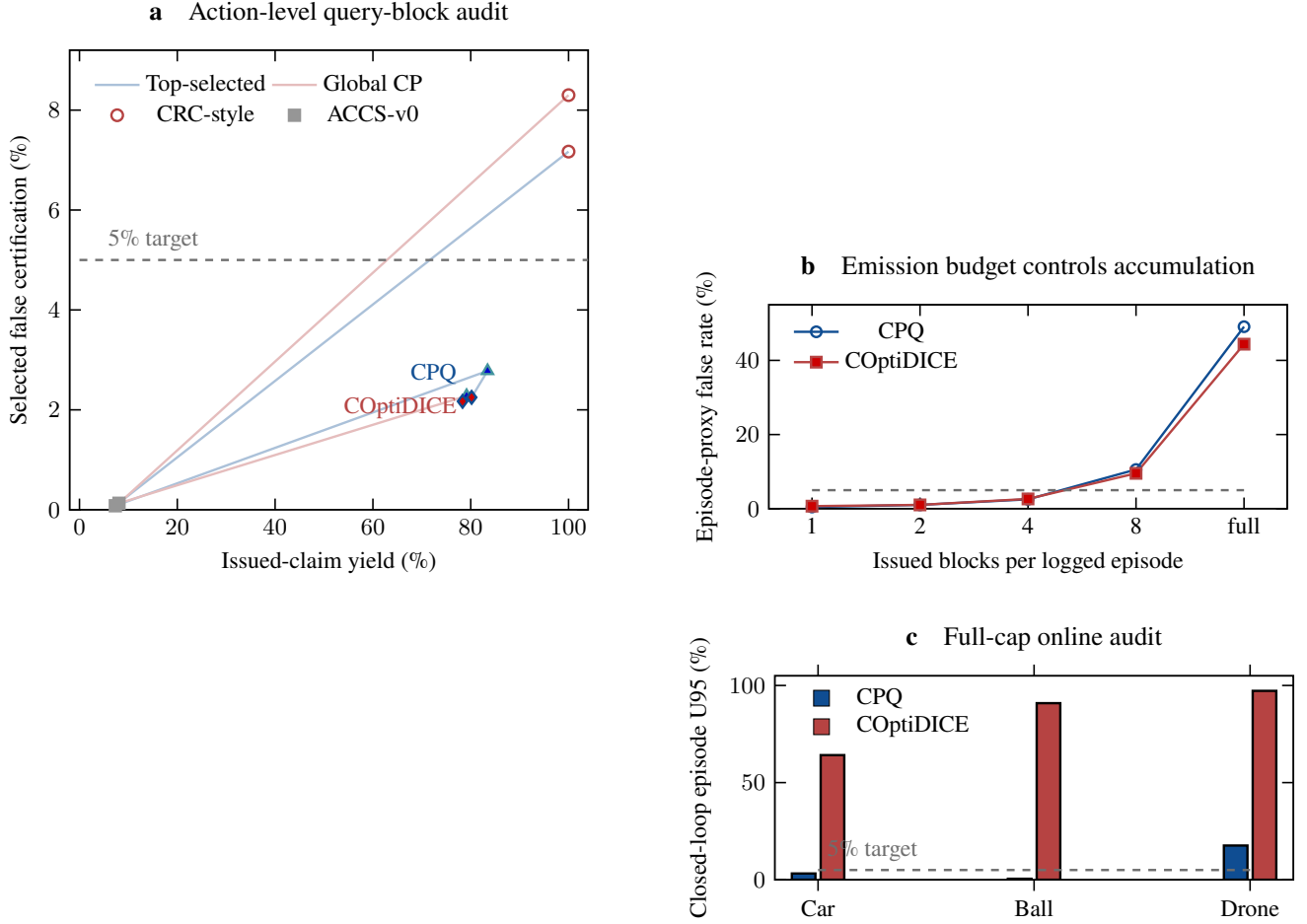


Figure 2: Three audit layers answer different safety questions. (a) On fixed CarCircle query blocks, ACCS-v0 moves the high-signal CPQ/COptiDICE rows below the 5% selected false-certification target while preserving much higher yield than global CP. (b) Grouping the same issued claims by logged episode shows why action-level control does not automatically imply episode-level control; sparse emission caps create an explicit risk-yield curve. (c) The pre-registered full-cap closed-loop audit gives exact episode-risk upper bounds: CPQ is below the 5% target on CarCircle and BallCircle, COptiDICE fails sharply, and DroneCircle exposes a harder residual-risk boundary for CPQ.

rect empirical comparison therefore requires runnable checkpoints or code that can emit the same fixed query blocks.

Reproducibility. Every promoted number is recomputable from saved query blocks or online audit traces. Action-level rows freeze checkpoints, candidate banks, support scores, residual labels, and splits before auditor comparison; episode-proxy rows group issued claims by logged episode; closed-loop rows use fixed rules and exact one-sided binomial bounds over issued online episodes.

Conclusion

This paper argues that a safety score becomes a certificate only when its deployment contract is auditable: query law, continuation label, support evidence, issued risk, yield, and episode budget. ACCS audits finite rule families on frozen query blocks and reports risk with yield; the evidence stack then separates action-level control on trained CarCir-

cle proposers, repeated-certificate diagnostics, and fixed-cap closed-loop certification for full CPQ CarCircle/BallCircle episodes. The failures are part of the result: low-yield global CP exposes a denominator problem, repaired toys justify trained RL query blocks, episode accumulation blocks trajectory overclaims, off-policy shift requires weighting or support assumptions, and COptiDICE/DroneCircle rejections prevent closed-loop overclaiming. Offline safe-RL deployment needs post-selection certificate evidence, not only safer proposers.

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